DISCRETE-TIME LOW-PASS FILTER DESIGN FOR DC MOTOR SPEED MEASUREMENT

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1.0 Learning Outcomes

After completing this exercise, you should be able to demonstrate the Learning Outcomes listed in Table 1. At the end of the exercise, revisit the Learning Outcomes in Table 1 and confirm that each has been achieved.

Table 1: *Learning Outcomes from the I Exercise*

ID	Learning Outcome	Achieved
1.	Implement a first-order discrete-time low-pass filter in MATLAB Simulink to smooth a measured signal.	☐
2.	Investigate the influence of filter coefficients on measurement variation and response delay.	☐
3.	Select, justify and verify suitable filter coefficients against defined performance requirements.	☐

2.0 Hardware Set-Up

This exercise builds on the hardware and algorithm setup described in the exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. Students are expected to develop and configure their own Simulink models to implement and evaluate a first-order discrete-time low-pass filter.

The investigation focuses on understanding how filter coefficients influence measurement variation and response delay. An experimental approach is adopted, allowing suitable coefficients to be selected through observation, measurement and comparison against defined performance requirements. No mathematical model of the DC motor is required.

Establish the Measurement and Requirements

1. Observe the unfiltered motor speed measurement while the motor operates at a constant speed. Capture the measurement data and examine the variation caused by measurement noise.

2. Establish a maximum acceptable steady-state measurement variation, using the standard deviation over a defined measurement interval. Select a realistic requirement based on the hardware and intended application.

3. Define a maximum acceptable response delay to ensure that the filtered signal responds sufficiently quickly to changes in motor speed.

Implement a Discrete-Time Filter

1. Implement a first-order discrete-time low-pass filter in Simulink using the difference equation:

$$y[k] = ax[k] + by[k-1]$$

   Select the coefficients such that $a = 1 - b$ to maintain unity steady-state gain, with $0 \leq b < 1$.

2. Set $b = 0$ and verify that the filtered output follows the input signal without any smoothing.

Investigate the filter coefficient

Repeat the previous test using the filter coefficients provided in Table 1. These values provide a structured starting point for investigating how the coefficients influence measurement smoothing and response delay. They are not intended to represent an optimal solution.

To ensure a fair comparison, use consistent operating conditions and measurement criteria throughout all trials. Quantify the steady-state measurement variation using the standard deviation of the measured signal over a defined interval. Evaluate the response delay by comparing the filtered and unfiltered speed measurements following a repeatable change in motor speed. Use the same response threshold and measurement method for every

trial to ensure a consistent comparison. For this exercise, maintain the same sampling interval throughout.

1. Evaluate each set of filter coefficients by measuring the steady-state variation while the motor operates at a constant speed, followed by the response delay when the motor speed changes. Record the results in Table 2.

2. Compare the experimental results against the previously established performance requirements. Identify which coefficients satisfy both the measurement variation and response delay criteria.

3. Select the most suitable filter coefficients based on the experimental evidence. If none of the suggested coefficients satisfies both requirements, refine the coefficient selection and repeat the investigation. Where both requirements cannot be achieved, explain the compromise between measurement smoothing and response delay.

Table 2: *Experimental Evaluation of Discrete-Time Low-Pass Filter Coefficients*

Trial	b	a = 1 − b	Steady-state variation	Response delay	Pass / fail
Reference	0.00	1.00			
A	0.50	0.50			
B	0.80	0.20			
C	0.90	0.10			
D	0.95	0.05			
Selected	_____	_____			

Final Filter Verification

1. Implement the selected filter coefficients and repeat the measurement tests at a different motor operating speed to investigate whether the filter maintains acceptable performance.

2. Verify that the filtered signal satisfies the specified steady-state measurement variation and response delay requirements.

3. Compare the final filtered and unfiltered measurement signals and justify the selected coefficients using the experimental evidence.

4.0 Concluding Remarks

This practical exercise introduced the implementation and experimental evaluation of a first-order discrete-time low-pass filter for reducing noise in DC motor speed measurements. Using MATLAB, Simulink and ACE-Lab hardware, the influence of different filter coefficients on measurement variation and response delay was investigated.

The investigation highlighted the compromise between measurement smoothing and response delay. Increasing the filter coefficient b generally improves the suppression of unwanted measurement variation but reduces the responsiveness of the filtered signal to changes in motor speed. Consequently, filter coefficients must be selected with consideration of the requirements of the intended application.

By evaluating experimental results against defined performance requirements, students can select, justify and verify suitable filter coefficients without relying on mathematical modelling. This reinforces the importance of evidence-based design and demonstrates how discrete-time filtering can improve measurement quality while maintaining an acceptable system response.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

> Pickering, J. E. (2026). Discrete-Time Low-Pass Filter Design for DC Motor Speed Measurement. ACE-Lab. Teaching Notes: ACE-Lab Practical Exercises, Version 1.0. Zenodo. https://zenodo.org/records/23244891